Lab : 06 (Git, Members 3 backend)

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Chapter 2: Parity Bit Notes.

=> what is parity bit: one extra bit added to data - it tells whether the number of 1s in the data is even or odd.

=> why it is used: when data is sent or stored, a bit can flip (0 becomes 1 or 1 becomes 0). Parity helps detect this error.

- Even Parity: chose the parity bit so that the total number of 1's is even.
- odd Parity: chose the parity bit so that the total number of 1s is odd.

-> Formula: Even parity bit = (count of 1s) % 2

odd parity bit = 1 - (count of 1s % 2)

-> Checker:- The receiver counts the 1's again. with even parity, an odd count means "error found". and an even count means "no error".

-> Limitations:- Parity detects only a single-bit

error (or an odd number of flipped bits).
if 2 bits flip, it is not detected. it can detect errors but cannot correct them.

Solved Example 1 (Generator):-

Data = 10110010

1- Count the 1s = 1+0+1+1+0+0+1+0 = 4

2- Even parity bit = 4 % 2 = 0, so the sent data is = 101100100.

3- Odd parity bit = 1-0 = 1, so the send data is 101100101.

Solved Example 2:- (Checker)

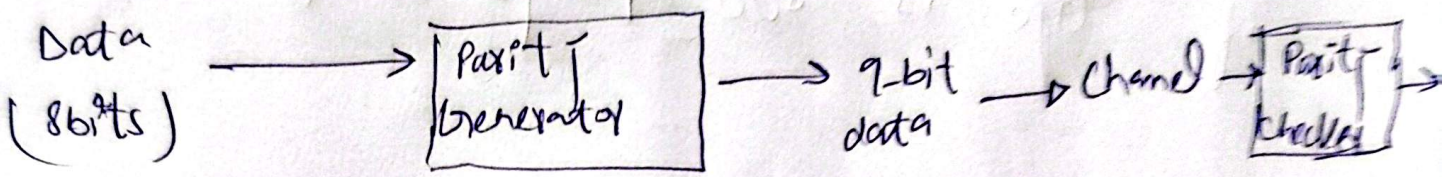
Even parity is used. The sender sent 101100100 but the receiver got 101100110 (the 8th bit flipped).

CamScanner

1- Count the 1s : 1+0+1+1+0+1+0+1+1+1+0 = 7.

2- 5 is odd, but even parity needs an even count.

3- Result : Error Found.



"No error / Error Found".

Step 2:- Pseudo - code

Function countOnes(bits)

1. Set count = 0.
2- Go through every character of the text.
3- if the character is '1', add to count.
4- Return count

Function createEvenParity(data, type)

1- if data is not exactly 8 characters of 0/1 → show error, STOP.

2- count = countOnes (data).

3- if type = "even": parity = count % 2.

if type is "odd": parity = 1 - (count % 2)

4- Final data = data + parity.

5- Show the count. the parity bit and the 9-bit data.

Function checkParity (received, type).

1- if received is not exactly 9 characters of 0/1 → show error, STOP.

2- count = countOnes (received).

3- Even parity: correct if count is even.
odd parity: correct if count is odd.

4. show "No error" or "Error found" with the count.

Function flipBit (Position)

1- Change that bit: 0 becomes 1, 1 becomes 0.
2- update the bit box on the page.